

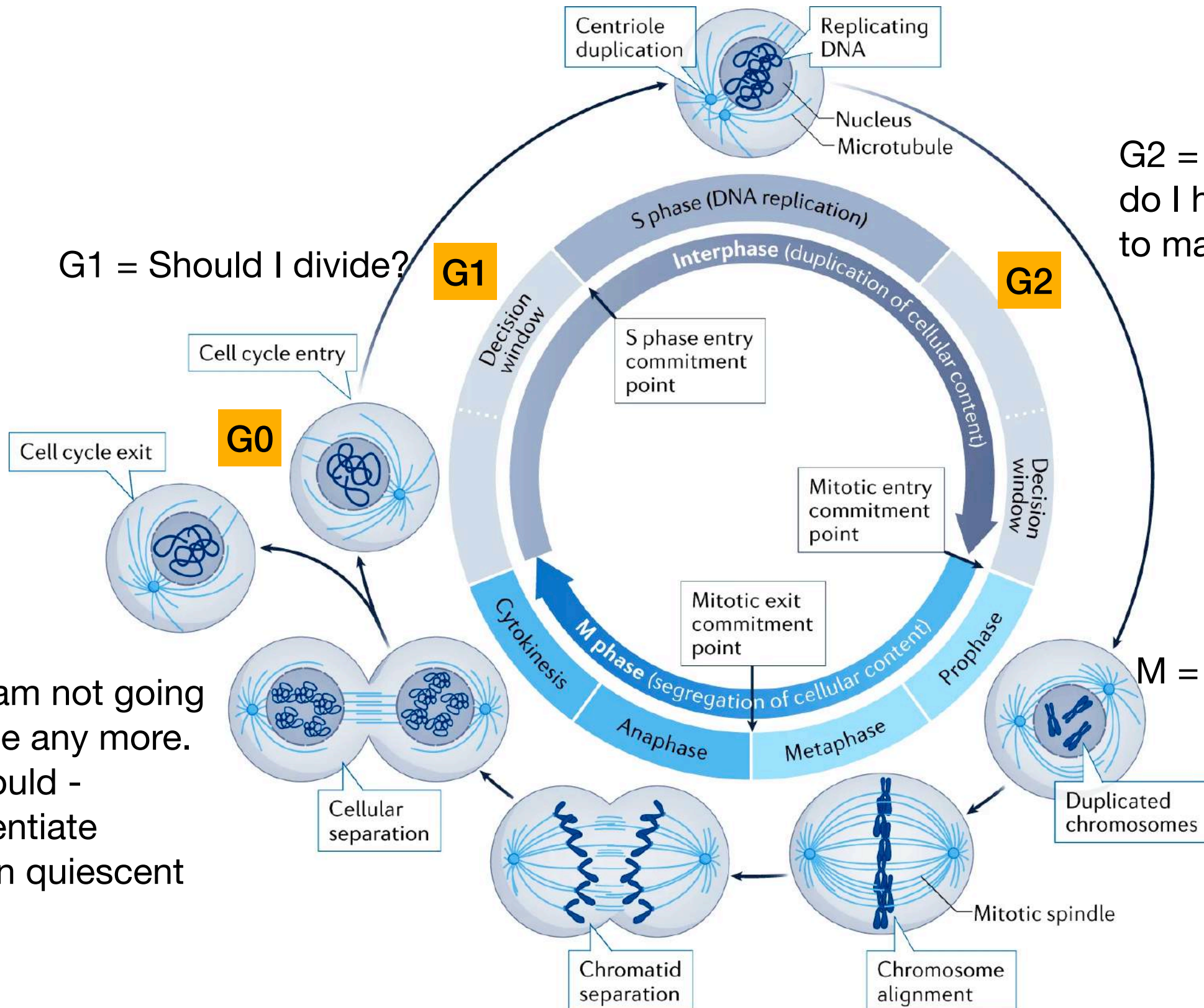
Cell Cycle

Cells that decide to continue to remain cycling through the cell cycle are capable of proliferation

S = I am doubling my DNA for daughter cells

G2 = In addition to DNA, do I have everything else to make 2 daughter cells?

G1 = Should I divide?



M = I am dividing into 2

I decided to divide and doubled my DNA. Have I ensured that the DNA will be allocated equally to both my daughters? = M phase = SAC

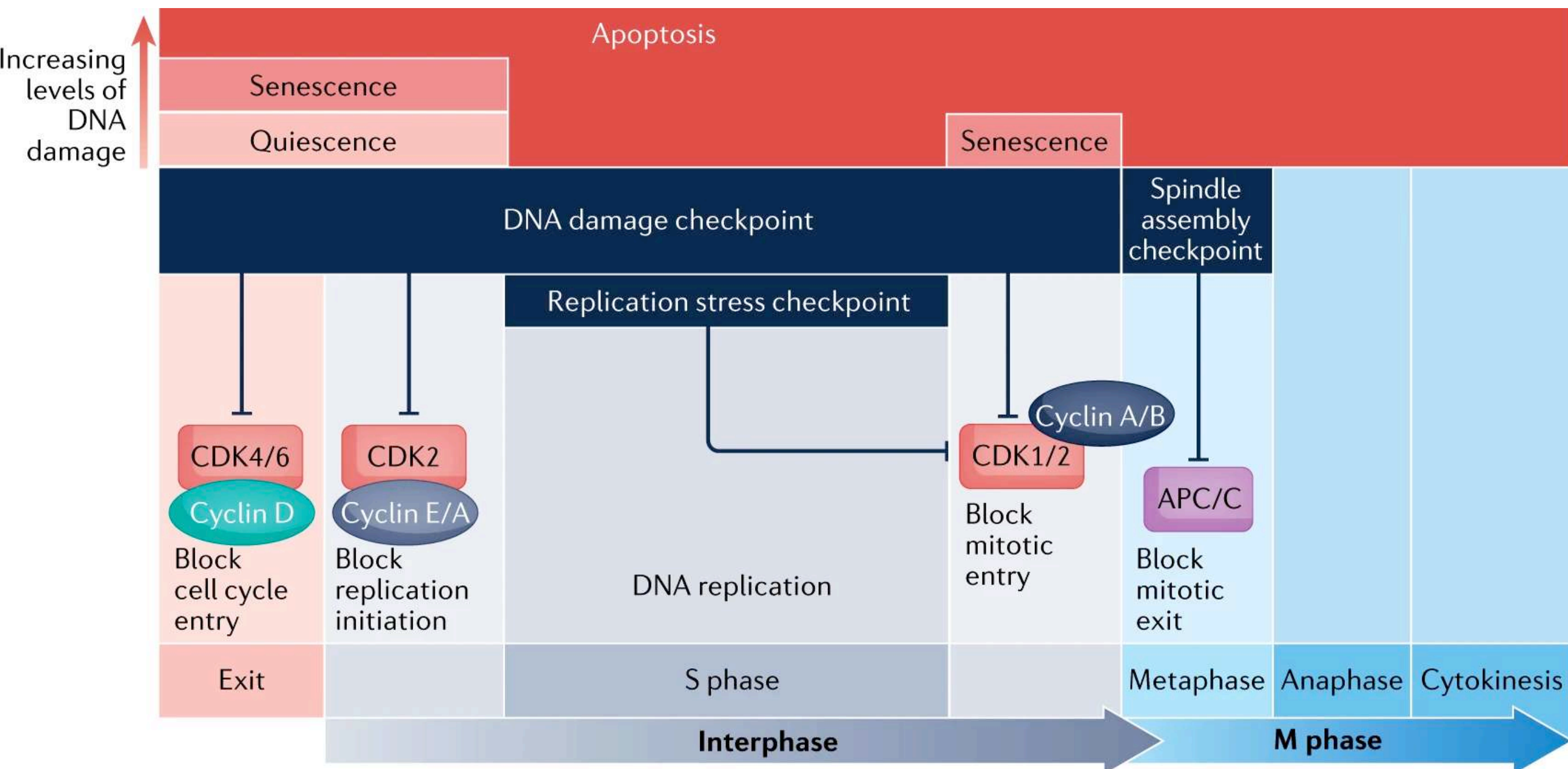
G0 = I am not going to divide any more. But I could -

- differentiate
- remain quiescent
- die

Cell Cycle

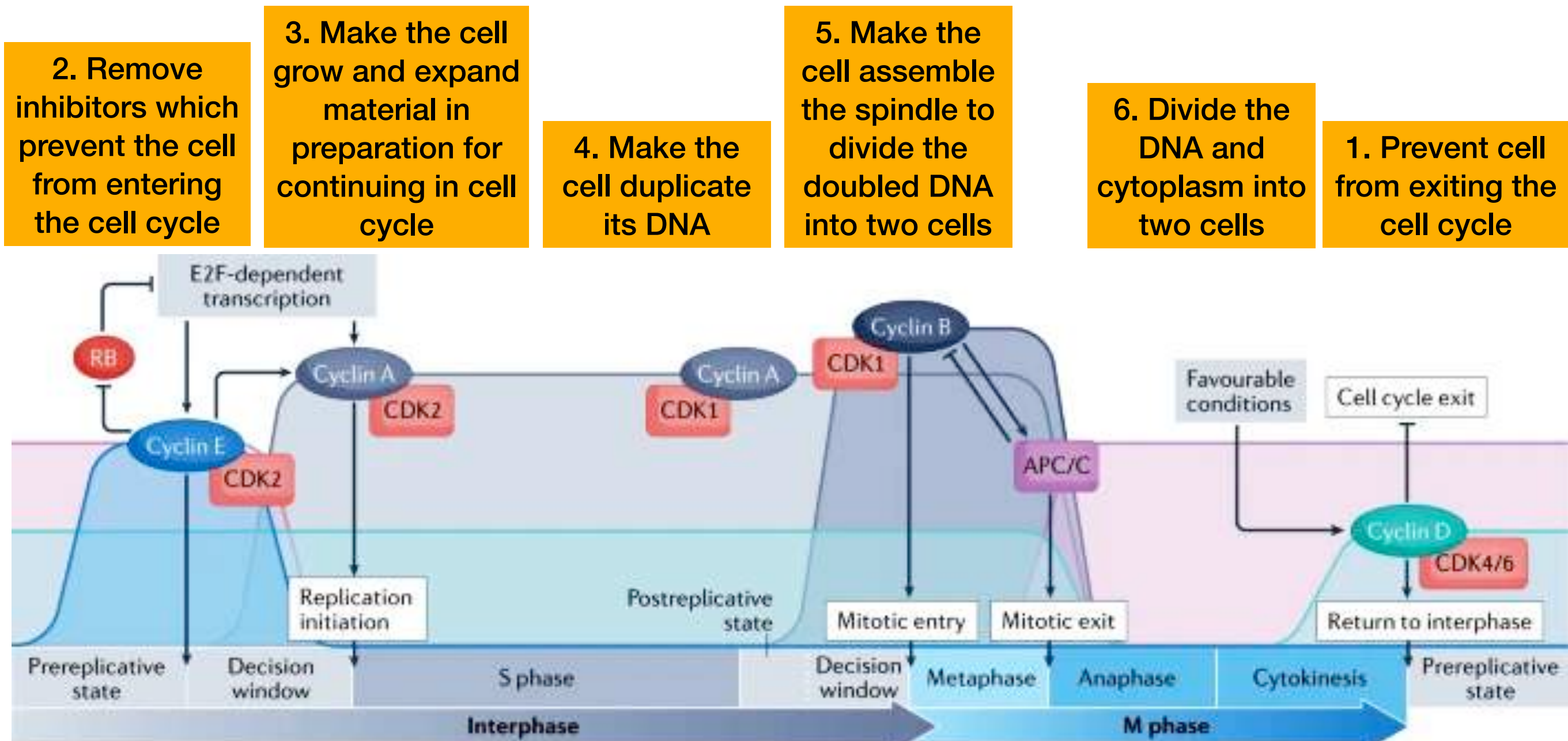
Progression through the cell cycle requires successfully “passing” several checkpoints

- Failure to clear a checkpoint results in stalling at that checkpoint till repair can finish to the satisfaction of passing the checkpoint
- Failure to repair and pass the checkpoint results in cell death



Progression through the cell cycle requires successfully “passing” the checkpoints

- Checkpoints = protein complexes of Cyclin Dependant Kinases (CDKs) and Cyclins
- Levels of Cyclin proteins rises and falls with phases of the cell cycle
- CDKs = enzymes present at all phases of the cell cycle, but not active unless partnered with a Cyclin
- Upon partnering with a Cyclin, the CDK-Cyclin pair phosphorylates other proteins
- The phosphorylated proteins get degraded allowing the cell to progress to the next phase and checkpoint



The cell cycle is an algorithm

- This algorithm has specific and logical steps to be followed in order to achieve specific outcomes of whether to divide or not
- This algorithm automatically checks for bugs in its code and stalls the algorithm till the bug gets fixed
- Failure to fix the bug results in failure of the algorithm and cell death = good outcome!
- Failure to detect the bug and/or failure to fix the bug results in the algorithm leading to unexpected outcomes = e.g. cancer = bad outcome!

2. Remove inhibitors which prevent the cell from entering the cell cycle

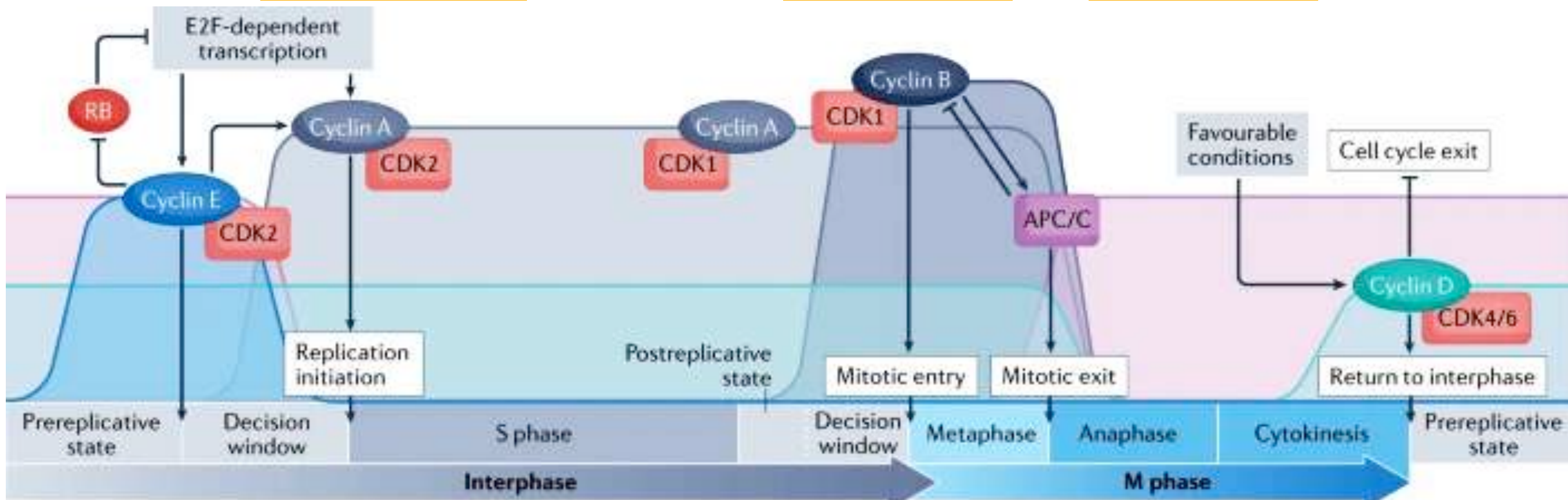
3. Make the cell grow and expand material in preparation for continuing in cell cycle

4. Make the cell duplicate its DNA

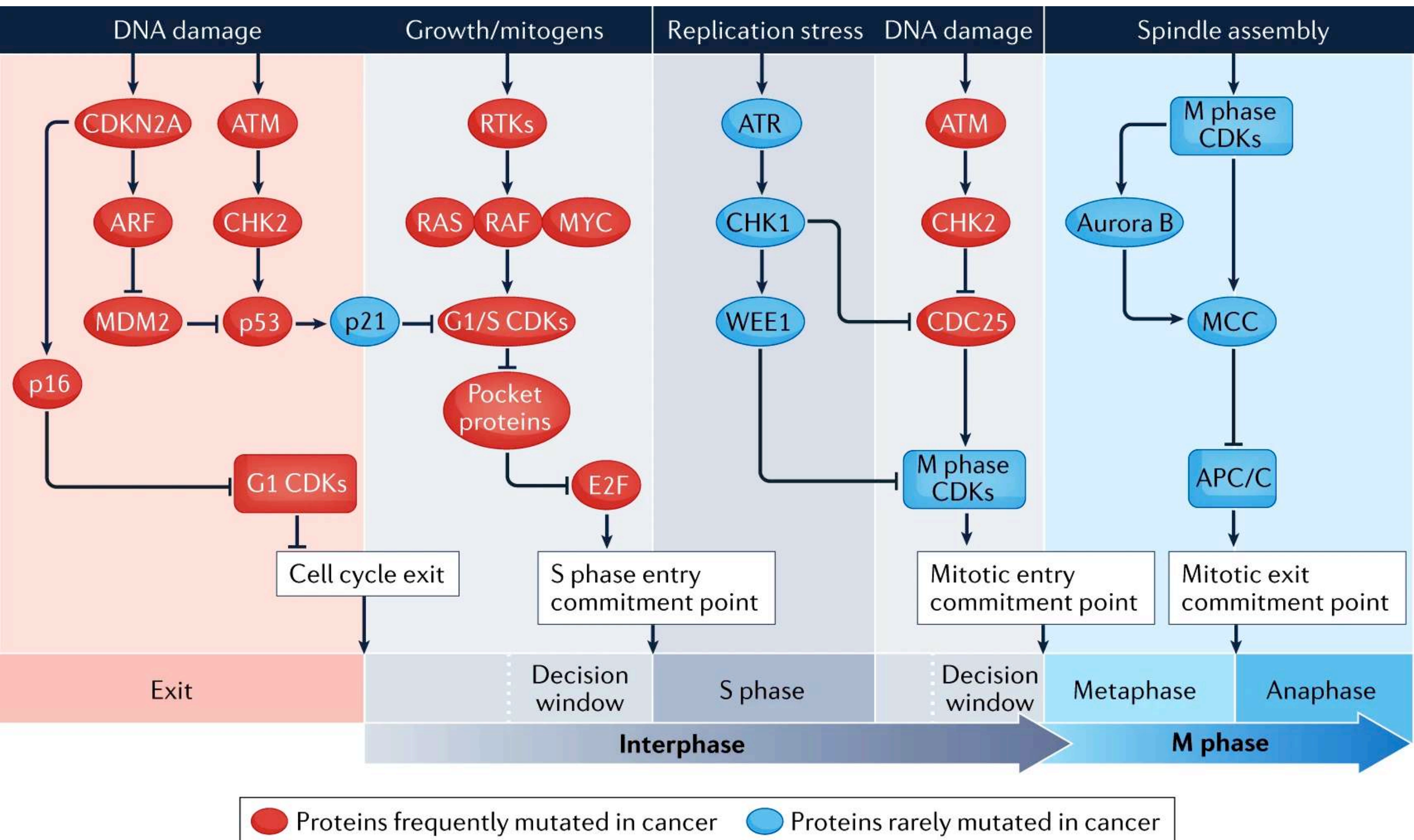
5. Make the cell assemble the spindle to divide the doubled DNA into two cells

6. Divide the DNA and cytoplasm into two cells

1. Prevent cell from exiting the cell cycle



Unfixable bugs in the cell cycle algorithm tend to occur more frequently in some parts of the code than in others



Unfixable bugs in the cell cycle algorithm are also opportunities!

fix the bug or alter the bug to get the algorithm to end without causing too much damage

Take advantage of the bug and make it go into overdrive to force the cell into a state of accelerated damage from where the only outcome is cell death

